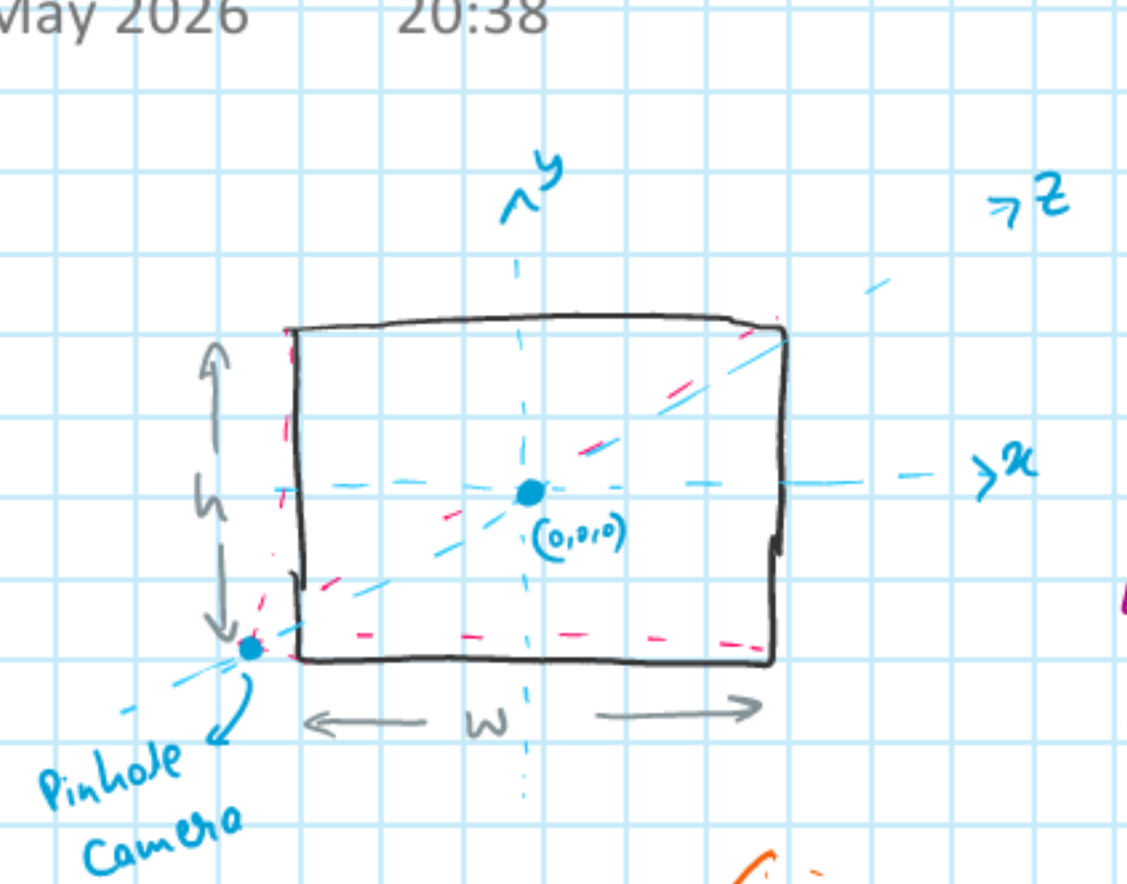


2D Relight

29 May 2026 20:38



camera $\rightarrow (0, 0, -\frac{\max(h,w)}{2 \tan(\frac{fov}{2})})$ $\xrightarrow{\text{user input}}$ field of view



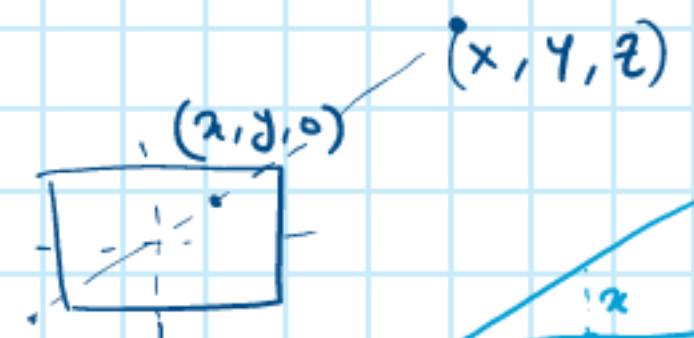
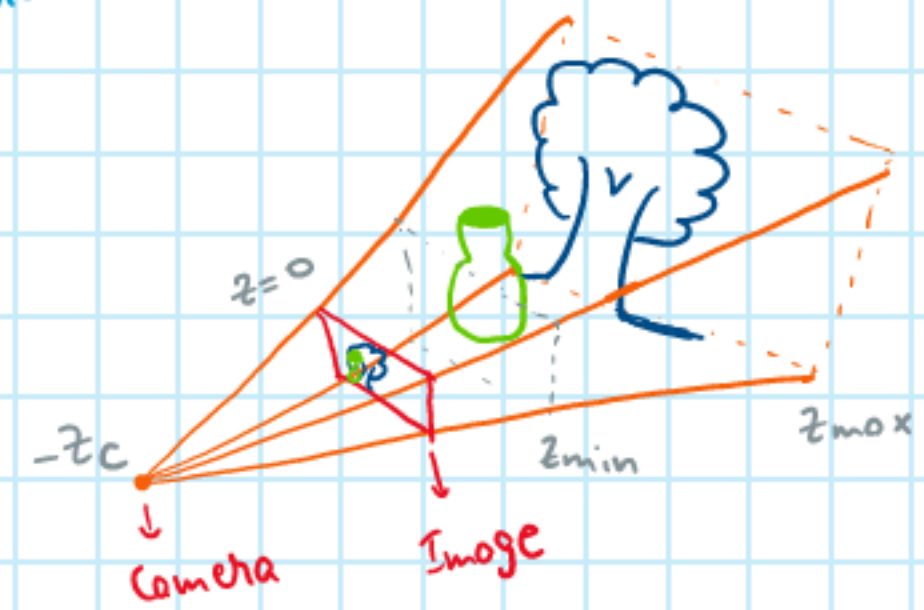
$$\tan\left(\frac{fov}{2}\right) = \frac{m/2}{z_c}$$



depth (x, y)
 $[0, 1]$

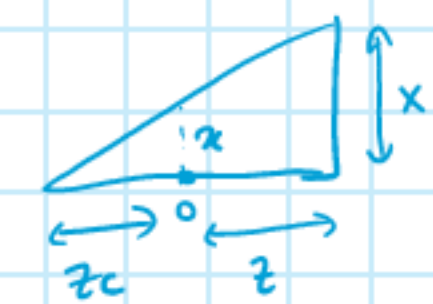
User Input

$$z_{min} = z_{min} \times z_c$$
$$z_{max} = z_{max} \times z_c$$



$$z = z_{min} + d(x, y) (z_{max} - z_{min})$$

given an (x, y) on image & $d(x, y)$, how to find (x, y, z) in 3D world



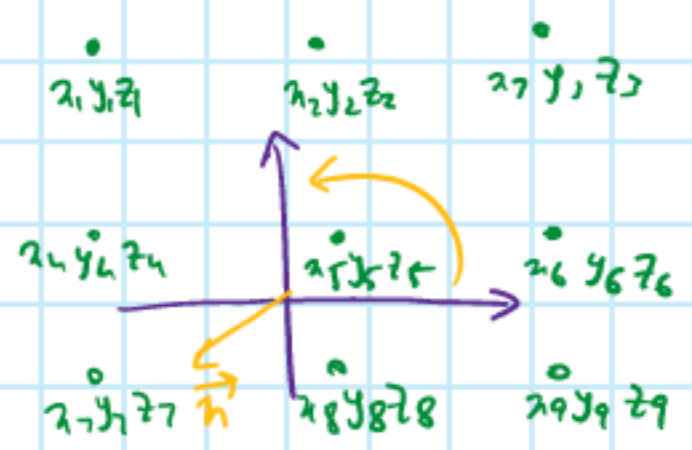
$$\frac{x}{z_c + z} = \frac{z}{z_c} \Rightarrow x = z \left(1 + \frac{z}{z_c}\right)$$

$$X = x \left(1 + z_{min} + d(x, y) [z_{max} - z_{min}]\right)$$
$$Y = y \left(1 + z_{min} + d(x, y) [z_{max} - z_{min}]\right)$$

$$\begin{matrix} z \\ y \\ d(x, y) \end{matrix} \xrightarrow{T} \begin{matrix} x \\ y \\ z \end{matrix}$$

So, now I know (x, y, z) of the scene in 3D

Estimating normal maps

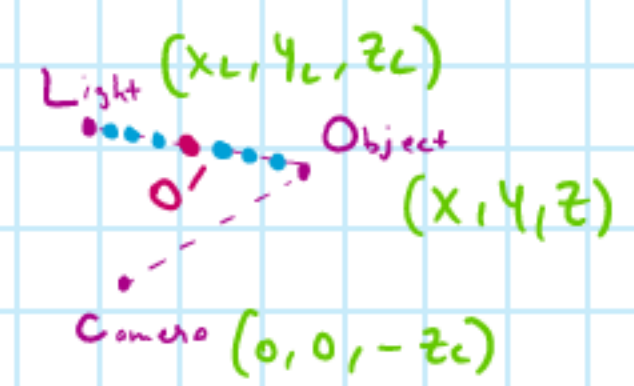
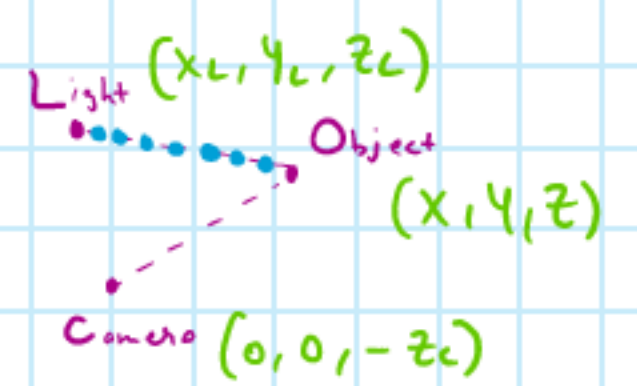
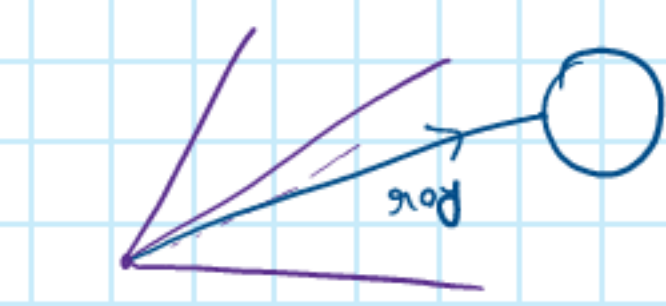


$$\vec{n} = (\vec{p}_6 - \vec{p}_4) \times (\vec{p}_2 - \vec{p}_8)$$

Light source @

$$(x_l, y_l, d(x, y))$$
$$\parallel$$
$$(x_l, y_l, z_l)$$

Ray Marching & PBR



for every (x, y, z) object pt,
for every (x', y', z') on Light - Object line,
 $x, y, d(x, y) = T^{-1}(x', y', z')$
if $z > z_{min} + d(x, y) [z_{max} - z_{min}]$:
Reject Ray $\Rightarrow$ shadow
Find BRDF and update pixel